WEEK 4 PROJECT

AI AGENT SYSTEMS

RAG using LangChain, LangGraph & FAISS

Presented by: Lokesh Thorat

1. Project Overview

What does the system do?

Builds a question-answering system using a supplied AI/ML notes file.

Retrieves relevant text before generating an answer.

Uses a local FLAN-T5 model for answer generation.

Connects the steps in a LangGraph workflow.

2. Problem Statement & Objectives

Problem

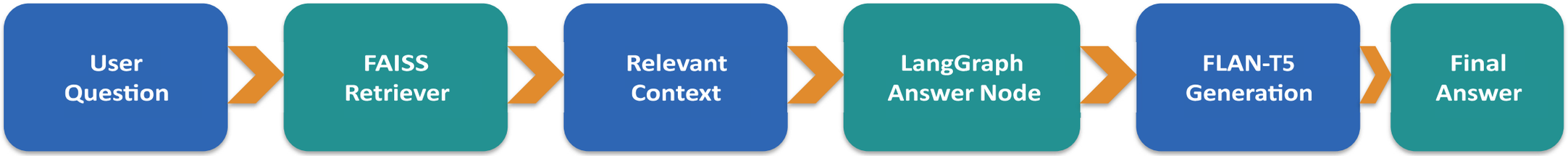
A model may answer without using the specific document supplied by the user.

Objectives

- Split notes into chunks
- Create text embeddings
- Retrieve relevant chunks with FAISS
- Generate a context-based response

3. System Architecture

RAG workflow



Source notes → text chunks → embeddings stored in FAISS → retrieve context → generate response

4. Tools & Technologies

LangChain

Document processing and retrieval components

FAISS

Stores vectors and finds similar text chunks

Sentence Transformers

Creates embeddings with all-MiniLM-L6-v2

LangGraph

Connects retrieval and answer-generation nodes

Hugging Face / FLAN-T5

Local text generation

Google Colab

Notebook execution environment

5. Implementation Steps

Prepared an AI/ML notes text file as the knowledge source.

Split text using chunk_size = 250 and chunk_overlap = 40.

Generated embeddings with all-MiniLM-L6-v2 and indexed them in FAISS.

Built a LangGraph workflow with retrieval and answer nodes.

Used a local FLAN-T5 model after the hosted API returned an insufficient-quota error.

6. Testing & Results

Six questions across AI/ML topics

AI

Definition returned

Machine Learning

Definition returned

Deep Learning

Definition returned

NLP

Rule-based fallback response

Computer Vision

Definition returned

AI Applications

Examples returned

6/6 questions returned non-empty answers (100% answer-generation rate; not accuracy).

7. Evaluation & Observations

Evaluation results were saved to rag_evaluation_results.csv.

Text similarity compares wording; it does not prove factual correctness.

The NLP answer uses a rule-based fallback because the local model output was too short.

LangSmith was configured, but traces were not confirmed in the dashboard.

8. Limitations & Future Work

Limitations

- Small knowledge base
- Small local model may give incomplete answers
- Only six test questions
- LangSmith traces remain unverified

Future Improvements

- Add more trusted documents
- Expand and review the test set
- Improve prompts and inspect retrieved chunks
- Verify tracing and add a user interface

CONCLUSION

The project demonstrates document chunking, embeddings, FAISS retrieval, and a LangGraph-based RAG workflow with local generation.

Thank You!!!!